INCH-POUND

A-A-54950
29 July 1994

COMMERCIAL ITEM DESCRIPTION

SUCTION AND PRESSURE APPARATUS, SURGICAL

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers two types of portable surgical suction and pressure apparatus.

Salient characteristics:

The surgical suction and pressure apparatus shall be of the following types:

Type I - With provisioning as specified in the solicitation/order. With electrical requirements as specified in the solicitation/order.

Type II - Without provisioning. Shall operate from 115/230 Volt, 50/60 Hz.

The suction and pressure apparatus shall be portable, self contained, electrically powered, and capable of long term constant suction. It shall be suitable for use in operating rooms with any of the power sources as specified herein, during surgical procedures, and during emergency care of the patient.

The apparatus shall have the following minimum characteristics:

Shall consist of at least a power cord, circuit breaker, electric on/off switch, primary voltage select switch, motor, pump with both suction and pressure connectible ports, vacuum regulator, vacuum gage, inline bacterial hydrophobic filter system, aspirate overflow protection, two drainage bottles, two sections of drainage tubing, all necessary "air path"

AMSC N/A FSC 6515

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interconnecting parts between the ambient air and patient catheter (not a component) to sustain the apparatus as a self-contained unit to be used as intended, operating/instruction manuals, all contained in a component carrying case, assembled and ready to use without being removed from carrying case.

Components shall have at least the following characteristics:

* Power cord - The power cord shall be at least 8 feet long and include a three-pronged, hospital-grade plug.

* Circuit breakers - The circuit breakers shall be properly rated to load and shall be in accordance with the appropriate standards.

* Electric on/off switch - An electric on/off switch shall be provided and shall be in accordance with the appropriate standards.

* Primary voltage switch - The apparatus shall be provided with an electric/electronic circuit that includes the capacity to operate from any power source as specified. A nonaccessible switch shall be provided to allow switching from and to 115 volts and 220 volts power source. The switch may be external if key operated and two keys are provided, or internal if tools are required to access the switch. Such internal location shall not be accessible during normal intended use or normal cleaning/sterilizing. For type I, this is in addition to other electrical requirements specified in the solicitation.

* Motor - The motor shall be rated at least 1/10 horsepower and shall be in accordance with the appropriate standards.

* Pump - The pump, when powered by the motor, shall be capable of producing 30 Lpm free airflow as measured at the distal end of a connected six foot drainage tube without the bacterial hydrophobic filter connected. The apparatus shall be capable of producing at least 24 Lpm free airflow as measured at the distal end of a connected six foot drainage tube with the bacterial hydrophobic filter connected. It shall be capable of maintaining negative pressure of twenty inches Hg (plus 5, minus 0 inches Hg), at zero airflow. The pump shall be capable of maintaining a positive pressure of 10 pounds per square inch (plus 5, minus 0 psi), at zero

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airflow. It shall be capable of operating sixteen hours continuously, under full rated load, without damage to the pump and/or motor, and without causing the motor to have a temperature rise in excess of its rated temperature rise rating. The positive pressure port shall be provided with a straight, tapered, serrated, tubing connector designed to accommodate tubing with inside diameters of 3/16, 1/4, 5/16, and 7/16 inch. The vacuum port shall be provided with a connector designed for quick disconnect/connect of drainage tubing.

* Vacuum regulator - The vacuum regulator shall be capable of continuously adjusting the negative pressure from 0 to the maximum pump capacity.

* Vacuum gage - The vacuum gage shall have a minimum diameter of two inches. Shall have appropriate dual Metric/English markings from 0 to 760 mm Hg/30 in. Hg. Each scale shall be separately color coded.

* Inline bacterial hydrophobic filter - The inline bacterial hydrophobic filter shall be capable of filtering the bacteria to prevent it from entering the pump.

* Aspirate overflow protection - The aspirate overflow protection shall protect the pump from damage by debris, secretions, and fluids in the aspirate overflow. May be combined with the filter described in the previous paragraph.

* Drainage bottles, two required - The two required drainage bottles shall be reusable, capable of withstanding a vacuum of 25 inches Hg without collapsing, and withstanding repeated gas sterilization. They shall be fabricated of clear, nonshattering material other than glass. Each bottle shall be graduated in cubic centimeters (cc) and have a capacity of at least 1600 cc with a usable capacity of at least 1400 cc. Tube fittings shall have an inside diameter of 7 mm.

* Drainage tubing, two required - Each section shall be sterile, single patient use, and be approximately six feet in length with an internal diameter of 7 mm. They shall be compatible with drainage bottle tube fittings. Drainage tubing shall have a wall thickness of at least 0.06 inches and shall be in compliance with ASTM F960 resistance to collapse at a vacuum level of 25 inches Hg.

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* "Air path" interconnecting parts - All parts that are in the aspirate path to, but not including, the filter shall be sterilizable via liquid sterilizing agent(s) and shall be capable of quick disassembly without use of tools. All parts shall be compatible with additional collection containers (not included) described in CID A-A-51602 (see Note 1 at end of this document). All components/parts in the "air-path" shall be in accordance with ASTM F960.

* Operation instruction, and service maintenance instruction manuals - Manuals for both type I and type II shall illustrate incorporation of additional collection bottles in conformance with Figures 1 and 2. Type I manual shall be as specified in the solicitation. Type II manual shall be in conformance with ASTM F960.

* Carrying case - The carrying case shall protect the unit during transport (emergency service and long distance shipments) and long term storage. It shall be provided with dust and water resistant gasket or interlocking seal and a carrying handle. Fabrication shall be of suitable semirigid plastic at least 1/8 inch thick. The carrying case design shall allow the apparatus to be used without being removed from the case (except the power cord and drainage tube). The case shall be equipped with exterior feet that restrict movement during use. The exterior dimensions shall be approximately 19 inches by 15 inches by 8 inches, and with apparatus contained therein, shall weigh approximately 35 pounds. General Requirements for plastics do not apply.

* Quick disconnect - The apparatus and collection canister shall be capable of being removed from its case. It shall be fitted with quick disconnect brackets that shall be in conformance with Figure 3 herein. The mounting bracket shall have a locking mechanism that will positively lock the suction and pressure apparatus and collection canisters to their respective mounting rails. In the locked position, the lock shall prevent horizontal movement of the apparatus or collection canisters along their respective mounting rails. In the unlocked position, the lock shall allow smooth travel of the apparatus or collection canisters along the rails. The lock shall be constructed to allow its operation (either locked or unlocked) using only one hand. The bracket and lock assembly shall be constructed in a manner that will prevent the apparatus and collection canisters from falling spontaneously in case of lock failure. To remove the suction apparatus or collection canisters from their respective mounting rails, a deliberate "up and out" motion only shall be required.

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The case shall have a length of the railing, in conformance with Figure 3, mounted in the lid and in the bottom of the case. These lengths of railing shall afford a means to lock the apparatus and collection canisters into the case. To facilitate removal from the case, the locks shall operate freely within the case without the use of tools. The time required to remove the apparatus and collection canisters from the case and remount it on the mounting rails shall not exceed one minute.

* Metal parts - All metal parts shall be fabricated of suitable corrosion-resisting metals or be suitably treated for corrosion-resistance. All metal parts shall be in accordance with the appropriate General Requirements.

* Plastic parts - All plastic parts shall be made of high impact plastic materials.

* Noise - The unit shall be quiet during operation under the most disadvantaged operating conditions within specified parameters, or the most disadvantaged conditions that the unit is intended to operate in accordance with the manufacturer's manual(s), whichever is greater. With the unit in normal attitude for operation, its noise level shall not produce more than 80 dBA at a floor distance of 180 cm from the unit as measured 90 cm and 180 cm above the floor. For test purpose, ambient sound level (unit not operating) shall be at least 10 dB less than the dB level with unit operating. During the test, there shall be no acoustically reflecting surface, other than ground (floor) within 10 feet of the unit. Weighting networks "a" in ASA 47 shall apply.

* Finish - The suction and pressure apparatus shall be fabricated from materials which are inherently resistant to corrosion or which are treated to render them corrosion-resistant. Where painting or plating is used, appropriate cleaning and priming shall be performed prior to finishing. Finish shall provide a pleasing professional appearance and be free of gaps, scratches, chips and other imperfections.

Workmanship. The surgical suction and pressure apparatus shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One surgical suction and pressure apparatus, as specified, constitutes one unit. Each unit shall be supplied in a suitable carton.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5 (percent defective) and an inspection level of S-2. The unit of product for examination purposes shall be one surgical suction and pressure apparatus.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5 (percent defective) and an inspection level of S-2. The unit of product for test purposes shall be one surgical suction and pressure apparatus.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Testing shall include applicable tests in ASTM F 960. Vacuum and free air flow tests shall be performed with all required and supplied elements comprising the collection system in place (i.e. tubing, filter as appropriate, dual collection canisters, overflow valves, etc.).

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTES:

1. Commercial Item Description (CID) A-A-51602 covers NSN: 6515-01-106-7644, Canister Assembly.

2. The following National Stock Numbers are covered by this document.

6515-01-377-1473 - Type I
6515-01-174-1477 - Type II

MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA - OSS

Project Number: 6515-5694

Location: SUCPR12C.DOC

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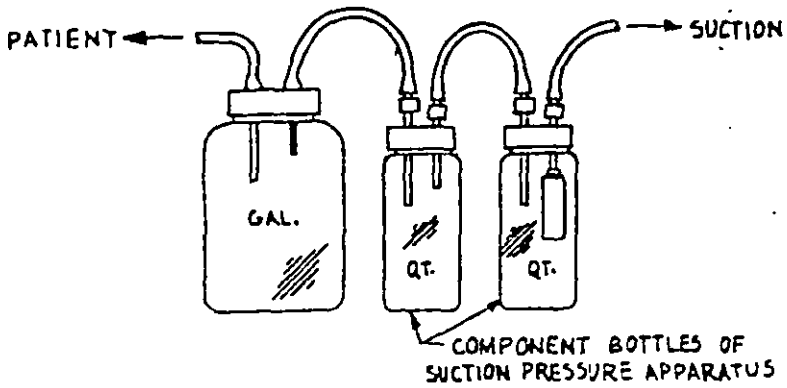


FIGURE 1

THREE SUCTION-BOTTLE SERIES

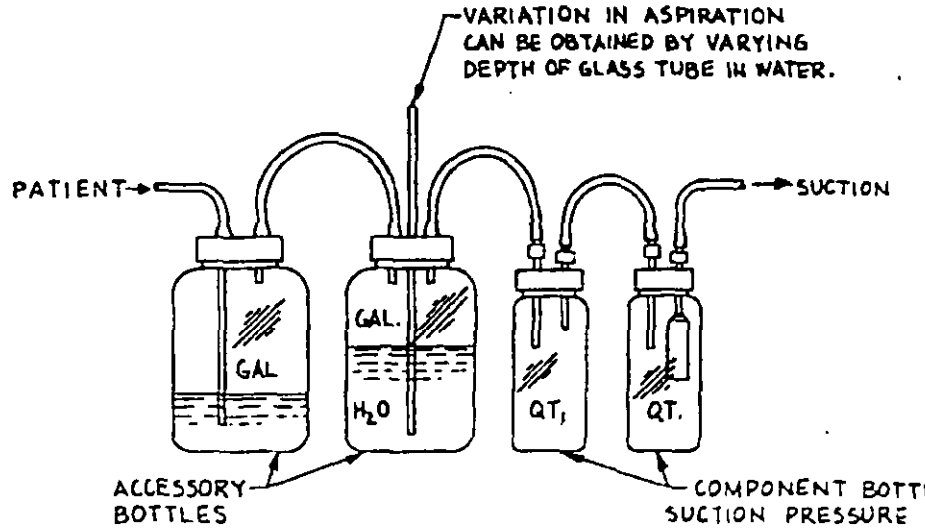
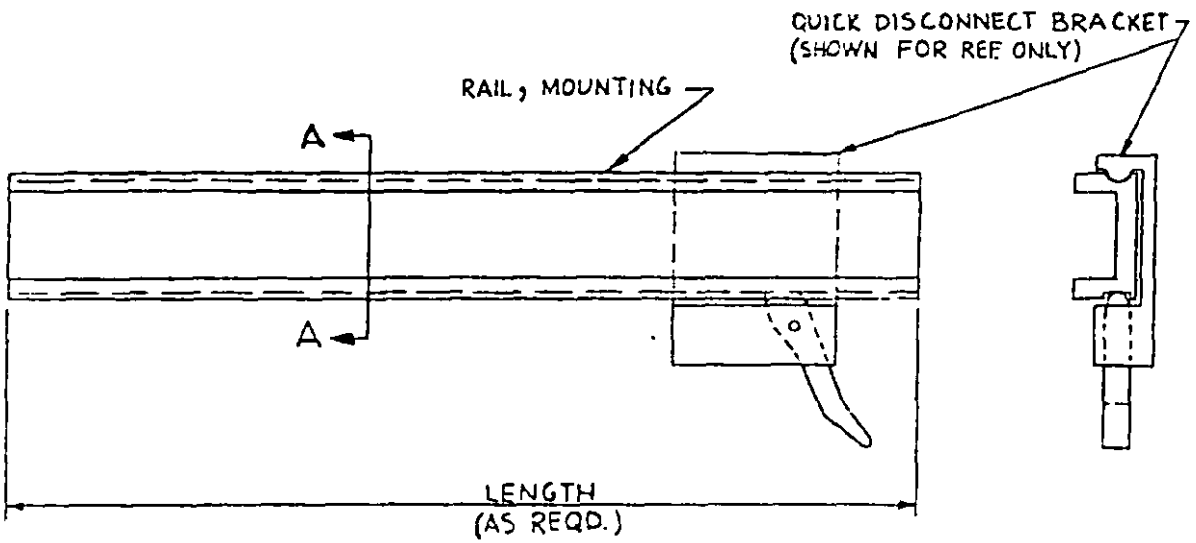


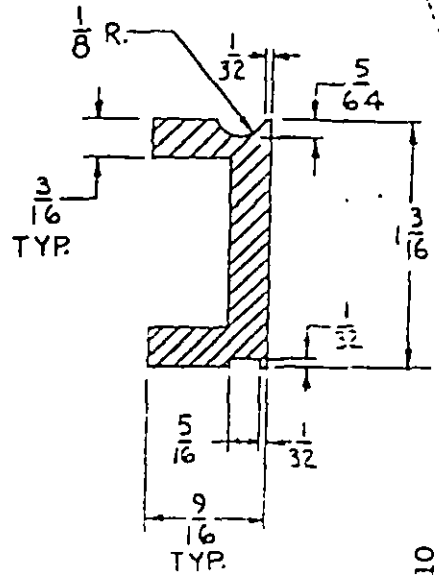
FIGURE 2

FOUR SUCTION-BOTTLE SERIES

NSN's 6515-01-377-1473
& 6515-01-174-1477

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SECTION A-A

SCALE: 2/1

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NOTES:

1. DIMENSIONS SHOWN ARE TO BE USED AS A GUIDE, AND MAY BE MODIFIED AS REQD TO PERMIT PROPER FUNCTIONING WITH QUICK DISCONNECT BRACKET.
2. THIS RAIL SHALL BE USED WITH A QUICK DISCONNECT BRACKET, A COMPONENT OF SUCTION AND PRESSURE APPARATUS NSN 6515-01-174-1477.
3. ALL DIMENSIONS ARE IN INCHES.

NSN's 6515-01-377-1473
& 6515-01-174-1477

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FIGURE 3 RAIL MOUNTING, FOR ATTACHMENT TO WALL.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54950	2. DOCUMENT DATE (YYMMDD) 940729

3. DOCUMENT TITLE
SUCTION AND PRESSURE APPARATUS, SURGICAL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MAP	b. TELEPHONE *(Include Area Code)* (1) Commercial (215)737-8431	(2) AUTOVON 444-8431
c. ADDRESS *(Include Zip Code)* 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete. 198/290

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